

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a01a5c46fd4c4a4ec.jsonl`  
- SHA-256: `0d7ff3e7b9855d53273dafacaa63f05111203dbd6a74eb2ce77a5fc714c05b2b`  
- Source modified: `2026-06-06T17:36:14+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:35:07.958Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Precise Event Spotting (PES), the regime needed for frame-accurate boundaries (e.g., shuttle contact / rally edges) in racket sports, requires detection thresholds of roughly 1-5 frames (~33-200 ms), far tighter than the 5-60 second/frame tolerance of general Action Spotting."

****Source:**** <https://arxiv.org/html/2505.03991v1> (primary)

****Supporting quote:**** "PES requires extremely strict detection thresholds, typically constrained within 1?5 frames (approximately 33?200 milliseconds)"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:35:10.370Z)

I'll verify this claim about Precise Event Spotting (PES) detection thresholds. Let me start by examining the primary source and searching for corroborating/contradicting evidence.

3. assistant (2026-06-06T17:35:13.383Z)

Now let me fetch the primary source and search for the relevant terminology and figures.

4. user (2026-06-06T17:35:28.552Z)

Analysis of PES vs Action Spotting Thresholds

****Paper Overview:****

This is a comprehensive survey titled "Action Spotting and Precise Event Detection in Sports: Datasets, Methods, and Challenges" covering deep learning approaches for sports video analysis.

****Domain:**** Sports analytics with focus on soccer, tennis, table tennis, and other sports. Does NOT focus on badminton specifically (though badminton datasets are mentioned).

Exact Quotes on PES Tolerance Thresholds

****Frame-level precision requirement:****

> "PES requires extremely strict detection thresholds, typically constrained within 1?5 frames (approximately 33?200 milliseconds at standard frame rates of 25?30 FPS)."

****Comparison between PES and AS tolerance windows:****

> "While AS facilitates coarser detection by tolerating errors within relatively broad temporal windows (typically between 5 and 60 seconds), certain applications in sports demand much higher

temporal precision."

****Critical event examples:****

> "even a margin of error of 1?2 frames can result in missing critical events, such as the precise moment of ball contact in tennis."

****Table 1 summary:****

The comparison table states PES tolerance is "0?2 frames" versus AS tolerance of "5?60 frames"

Key Distinction

The paper explicitly contrasts the two approaches: Action Spotting uses broad windows measured in seconds (5-60s), while Precise Event Spotting requires frame-accurate detection within 1-5 frames (~33-200ms).

5. user (2026-06-06T17:35:28.792Z)

Web search results for query: "Precise Event Spotting PES strict detection threshold 1-5 frames temporal tolerance action spotting"

Links: [{"title": "Action Spotting and Precise Event Detection in Sports: Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v1"}, {"title": "Few-Shot Precise Event Spotting via Unified Multi-Entity Graph and Distillation", "url": "https://arxiv.org/html/2511.14186"}, {"title": "Action Spotting and Precise Event Detection in Sports", "url": "https://arxiv.org/html/2505.03991"}, {"title": "Few-Shot Precise Event Spotting via Unified Multi-Entity Graph and Distillation", "url": "https://arxiv.org/pdf/2511.14186"}, {"title": "(PDF) Few-Shot Precise Event Spotting via Unified Multi-Entity Graph and Distillation", "url": "https://www.researchgate.net/publication/397642798_Few-Shot_Precise_Event_Spotting_via_Unified_Multi-Entity_Graph_and_Distillation"}, {"title": "Multi-Focus Temporal Shifting for Precise Event Spotting in Sports Videos", "url": "https://arxiv.org/html/2507.07381"}, {"title": "Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges", "url": "https://arxiv.org/pdf/2505.03991"}, {"title": "Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v3"}, {"title": "Precise Event Spotting in Sports Videos: Solving Long-Range Dependency and Class Imbalance", "url": "https://arxiv.org/pdf/2503.00147"}]

API Error: Internal server error

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:35:34.017Z)

The primary source confirms the quote verbatim. Let me cross-check independent literature to see if the 1-5 frame / PES vs the seconds-tolerance Action Spotting distinction is corroborated elsewhere (the foundational PES papers and SoccerNet conventions).

7. user (2026-06-06T17:35:41.995Z)

Web search results for query: "Precise Event Spotting E2E-Spot tennis table tennis evaluation tolerance 1 frame 2 frames metric"

Links: [{"title": "Multi-Focus Temporal Shifting for Precise Event Spotting in Sports Videos", "url": "https://arxiv.org/html/2507.07381"}, {"title": "Few-Shot Precise Event Spotting via Unified Multi-Entity Graph and Distillation", "url": "https://arxiv.org/html/2511.14186"}, {"title": "Precise Event Spotting in Sports Videos: Solving Long-Range Dependency and Class Imbalance", "url": "https://arxiv.org/pdf/2503.00147"}, {"title": "[Quick Review] Precise Event"}]

Spotting in Sports Videos: Solving Long-Range Dependency and Class Imbalance", "url": "https://liner.com/review/precise-event-spotting-in-sports-videos-solving-longrange-dependency-and"}, {"title": "Spotting Temporally Precise, Fine-Grained Events in Video", "url": "https://arxiv.org/pdf/2207.10213"}, {"title": "Precise Event Spotting in Sports Videos: Solving Long-Range Dependency and Class Imbalance", "url": "https://arxiv.org/html/2503.00147"}, {"title": "Action Spotting and Precise Event Detection in Sports: Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v1"}, {"title": "Spotting Temporally Precise, Fine-Grained Events in Video", "url": "https://jhong93.github.io/pdf/spot-suppl.pdf"}]

Based on the search results, here's information about Precise Event Spotting, E2E-Spot, and the evaluation metrics used for tennis and table tennis:

E2E-Spot Framework

E2E-Spot is an end-to-end deep learning framework for temporally precise spotting of fine-grained events in video, defined as predicting the exact frame (within 172 frames tolerance) when an event occurs. E2E-Spot employs a CNN backbone with Gated Shift Modules and GRUs for temporal modeling, and performed effectively in tennis and figure skating scenarios.

Evaluation Metrics and Tolerances

For the Tennis and Figure Skating datasets, which have very strict evaluation criteria, results are reported with $\tau=0, 1, \text{ and } 2$. The metric used is average precision (AP) within a specified tolerance range, denoted as τ , with the AP calculated for each event class to calculate the mean average precision (mAP) with the specified τ value as an evaluation metric.

Table Tennis Context

The Table Tennis Australia dataset is the first PES benchmark for table tennis containing over 4,800 precisely annotated events. Frame-level precision is critical in table tennis, where even minor inaccuracies can significantly impact the determination of ball bounce locations, given the rapid motion of the ball across frames in professional-level matches.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-06T17:35:51.347Z)

Web search results for query: "SoccerNet action spotting evaluation tolerance delta seconds 5 60 average-mAP"

Links: [{"title": "[1804.04527] SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos", "url": "https://arxiv.org/html/1804.04527"}, {"title": "[1804.04527v2] SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos", "url": "https://arxiv.org/abs/1804.04527v2"}, {"title": "SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos | Papers With Code", "url": "https://paperswithcode.com/paper/soccernet-a-scalable-dataset-for-action"}, {"title": "SoccerNet Dataset: Sports Video Benchmark", "url": "https://www.emergentmind.com/topics/soccernet-dataset"}, {"title": "SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos", "url": "https://arxiv.org/pdf/1804.04527"}, {"title": "Unifying Global and Local Scene Entities Modelling for Precise Action Spotting", "url": "https://arxiv.org/pdf/2404.09951"}, {"title": "SoccerNet", "url": "https://silviogiancola.github.io/SoccerNet/"}, {"title": "A Context-Aware Loss Function for Action Spotting in Soccer Videos", "url": "https://arxiv.org/pdf/1912.01326"}, {"title": "SoccerNet-v2: A Dataset and Benchmarks for Holistic Understanding of Broadcast Soccer Videos", "url": "https://arxiv.org/pdf/2011.13367"}, {"title": "(PDF) SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos", "url": "https://www.researchgate.net/publication/329751327_SoccerNet_A_Scalable_Dataset_for_Action_Spotting_in_Soccer_Videos"}]

Based on the search results, here's information about SoccerNet's action spotting evaluation

with the tolerance delta (?) range of 5 to 60 seconds and Average-mAP metric:

Evaluation Metric Definition

A predicted action spot is positive if it falls within a given tolerance ? of a ground-truth timestamp from the same class. The Average Precision (AP) based on PR curves is computed then averaged over the classes (mAP), after which the Average-mAP is the AUC of the mAP computed at different tolerances ? ranging from 5 to 60 seconds.

Baseline Performance

The baseline reaches an Average-mAP of 49.7% for tolerances ? ranging from 5 to 60 seconds.

Historical Context

SoccerNet-v1 evaluated action-spotting models on loose tolerance from 5 seconds to 60 seconds, which is called loose A-mAP. However, a new metric - the tight A-mAP (T-mAP) was introduced in SoccerNet-v2 with tight tolerance just ranging from 1 second to 5 seconds.

This evaluation framework allows researchers to assess action spotting performance across different temporal tolerance levels, with the Average-mAP providing a comprehensive score across the entire tolerance range.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:36:04.807Z)

I now have strong corroboration from multiple independent sources. Let me synthesize the verification.

The claim is well-supported and corroborated:

1. ****Quote support****: The primary source (arxiv 2505.03991v1, a 2025 survey "Action Spotting and Precise Event Detection in Sports") contains the exact supporting quote verbatim: "PES requires extremely strict detection thresholds, typically constrained within 1?5 frames (approximately 33?200 milliseconds at standard frame rates of 25?30 FPS)." The math checks out: at 30 FPS, 1 frame = 33ms, 5 frames = 167ms; at 25 FPS, 5 frames = 200ms. The claim accurately paraphrases the source.

2. ****The AS contrast is also directly supported****: The same paper states AS "tolerating errors within relatively broad temporal windows (typically between 5 and 60 seconds)." The claim's "5-60 second" tolerance for general Action Spotting matches.

3. ****Independent corroboration****: The foundational PES paper "Spotting Temporally Precise, Fine-Grained Events in Video" (Hong et al., E2E-Spot, arxiv 2207.10213) reports evaluation at ?=0, 1, 2 frames for tennis/figure skating ? confirming the 1-2 frame regime. SoccerNet (arxiv 1804.04527), the canonical Action Spotting benchmark, uses ? from 5 to 60 seconds ? confirming the "5-60 second" AS figure.

4. ****Currency****: Source is from 2025 (very current). The cited supporting works (E2E-Spot 2022, SoccerNet 2018) are the field-standard references.

5. ****Source quality****: This is a peer-style academic survey, not marketing/press release/forum. The claim's strength (a definitional/methodological statement) is well-matched to an academic survey, and the underlying conventions are independently verifiable.

One minor nuance: the claim says "5-60 second/frame tolerance" ? the survey's Table 1 apparently lists AS tolerance as "5-60 frames" while its prose says "5-60 seconds." This is an internal inconsistency in the source, but the prose (5-60 seconds) matches SoccerNet's actual convention, and the claim correctly used the standard "seconds" figure. This does not refute the claim; it actually shows the claim picked the correct figure.

The claim is accurate, well-sourced, current, and independently corroborated. I could not refute it.

10. user (2026-06-06T17:36:12.322Z)

Structured output provided successfully

11. assistant (2026-06-06T17:36:14.784Z)

I've completed the verification. The claim is ****not refuted**** ? it is well-supported, current, and independently corroborated. I submitted my findings via StructuredOutput with high confidence.